A Ruby application (on Rails or not), can leak memory - either in the Ruby code or at the C code level.

In this section, you will learn how to find and fix such leaks by using tool such as Valgrind.

5.1 Valgrind

Valgrind is a Linux-only application for detecting C-based memory leaks and race conditions.
There are Valgrind tools that can automatically detect many memory management and threading bugs, and profile your programs in detail. For example, if a C extension in the interpreter calls `malloc()` but doesn't properly call `free()`, this memory won't be available until the app terminates.
For further information on how to install Valgrind and use with Ruby, refer to Valgrind and Ruby by Evan Weaver.

6 Plugins for Debugging

There are some Rails plugins to help you to find errors and debug your application. Here is a list of useful plugins for debugging:

- Footnotes Every Rails page has footnotes that give request information and link back to your source via TextMate.
- Query Trace Adds query origin tracing to your logs.
- Query Reviewer This rails plugin not only runs "EXPLAIN" before each of your select queries in development, but provides a small DIV in the rendered output of each page with the summary of warnings for each query that it analyzed.
- Exception Notifier Provides a mailer object and a default set of templates for sending email notifications when errors occur in a Rails application.
- Better Errors Replaces the standard Rails error page with a new one containing more contextual information, like source code and variable inspection.
- RailsPanel Chrome extension for Rails development that will end your tailing of development.log. Have all information about your Rails app requests in the browser - in the Developer Tools panel. Provides insight to db/rendering/total times, parameter list, rendered views and more.

7 References

- ruby-debug Homepage
- debugger Homepage
- byebug Homepage
- Article: Debugging a Rails application with ruby-debug
- Ryan Bates' debugging ruby (revised) screencast
- Ryan Bates' stack trace screencast
- Ryan Bates' logger screencast
- Debugging with ruby-debug